

FLUX

A VISUAL DATA PIPELINE BUILDER

SPL-II: SWE 4404

PROJECT PROPOSAL

TEAM 5

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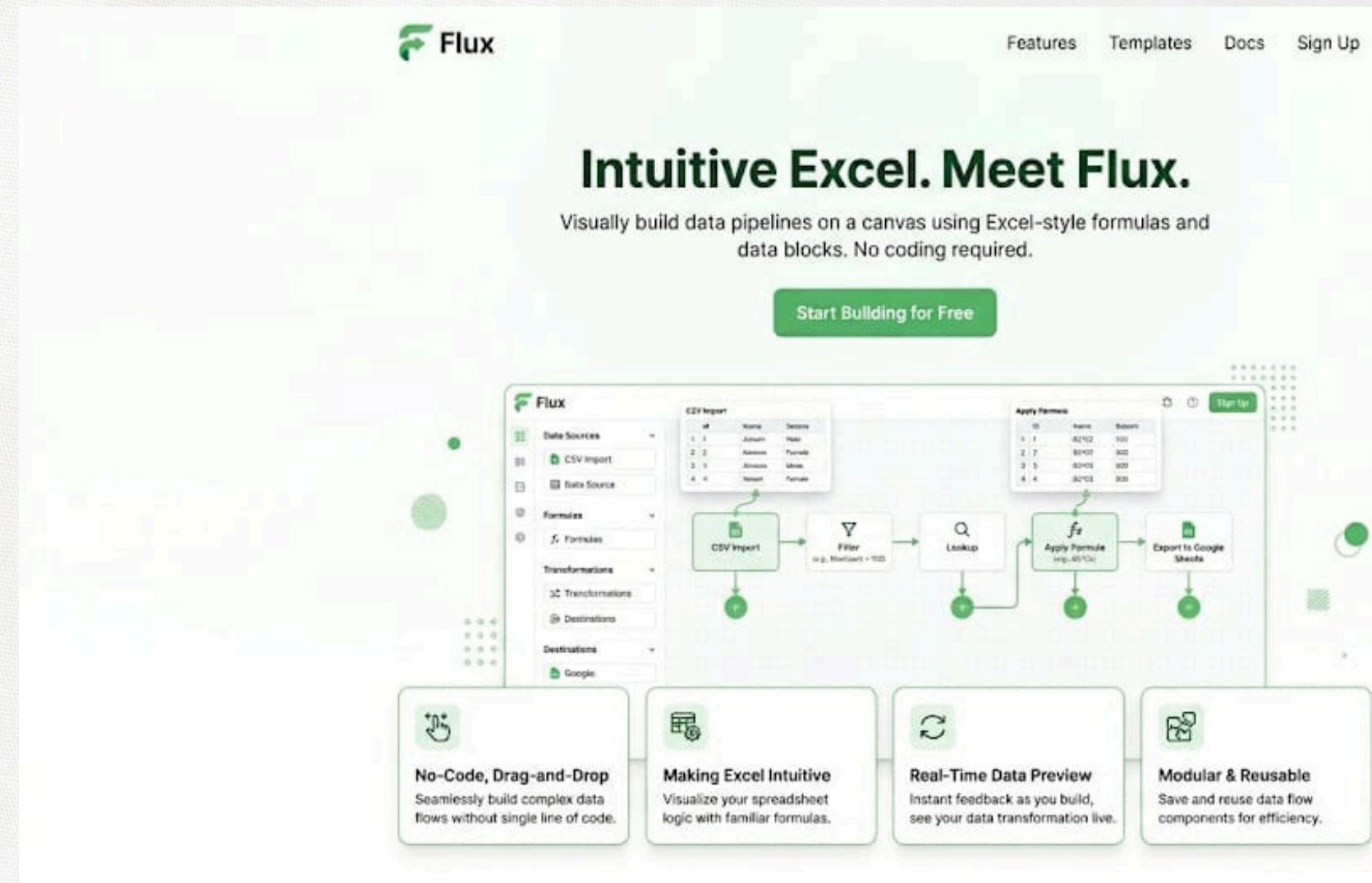
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OVERVIEW

- No-code, drag-and-drop application
- Visual data pipeline builder
- Real Time data preview
- Modular & reusable dataflow
- Making excel intuitive
- Reverse engineering formula's



MOTIVATION

- Reduce reliance on complex formulas
- Empower non-technical users to handle data
- Prevent workflows from breaking on data change
- Make debugging faster and easier
- Improve transparency in data processing

KEY FEATURES

➤ Drag-and-drop canvas — no code

- Drag blocks from sidebar
- Connect to build pipeline

➤ Live previews after every block

- Real-time table updates
- See transformations instantly

➤ Save pipeline, swap data, rerun

- Save as reusable template
- Change source file only

➤ Build pipeline → Excel formula

- Formula generated live
- Export working .csv files

KEY FEATURES

➤ Paste formula → visual pipeline

- Reverse engineer Excel formulas
- See logic as blocks

➤ Step-by-step replay animation

- Play data flow animation
- Spot errors instantly

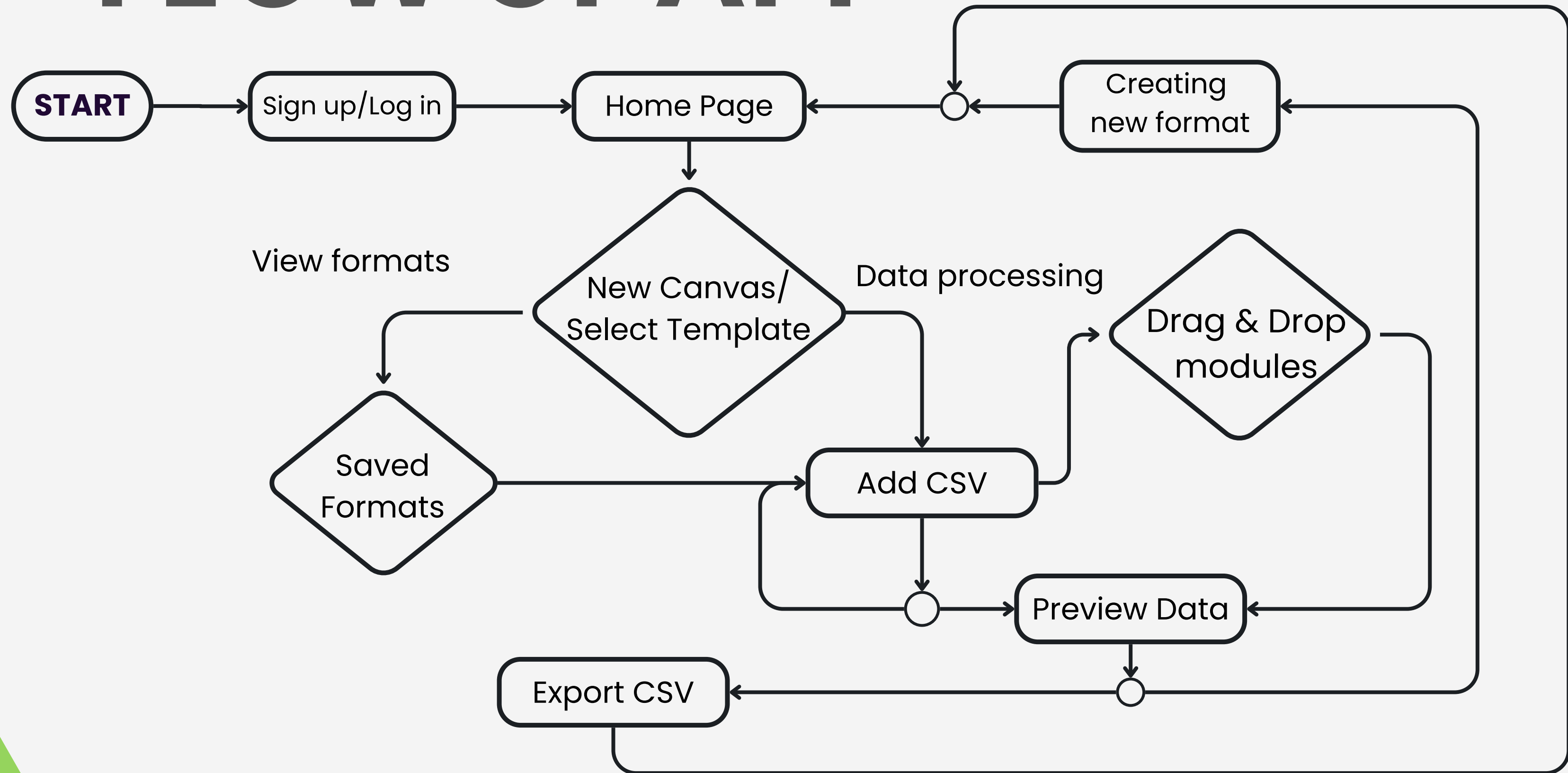
➤ Error blocks highlighted instantly

- Red glow on faulty block
- Affected rows flagged

USE CASE

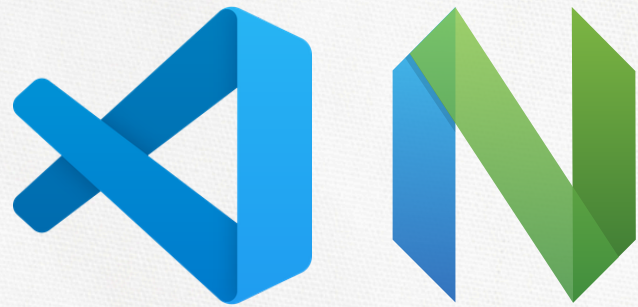
- Data cleaning and preprocessing for analysis
- Business reporting and dashboard preparation
- Academic and research data analysis
- Financial data processing and budgeting
- ETL (Extract, Transform, Load) workflows
- Quick prototyping of data pipelines
- Teaching and learning data concepts visually
- Automating repetitive spreadsheet tasks

FLOW OF APP



TOOLS & TECHNOLOGY

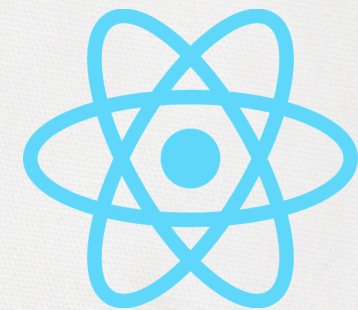
IDE



BACKEND



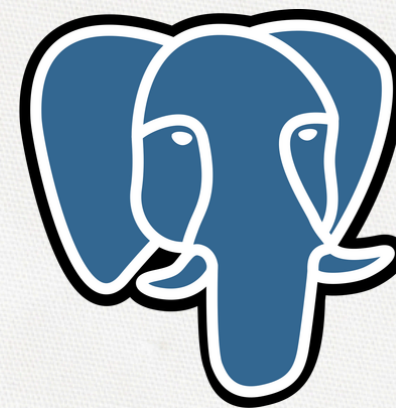
FRONTEND



DOCUMENTATION



DATABASE

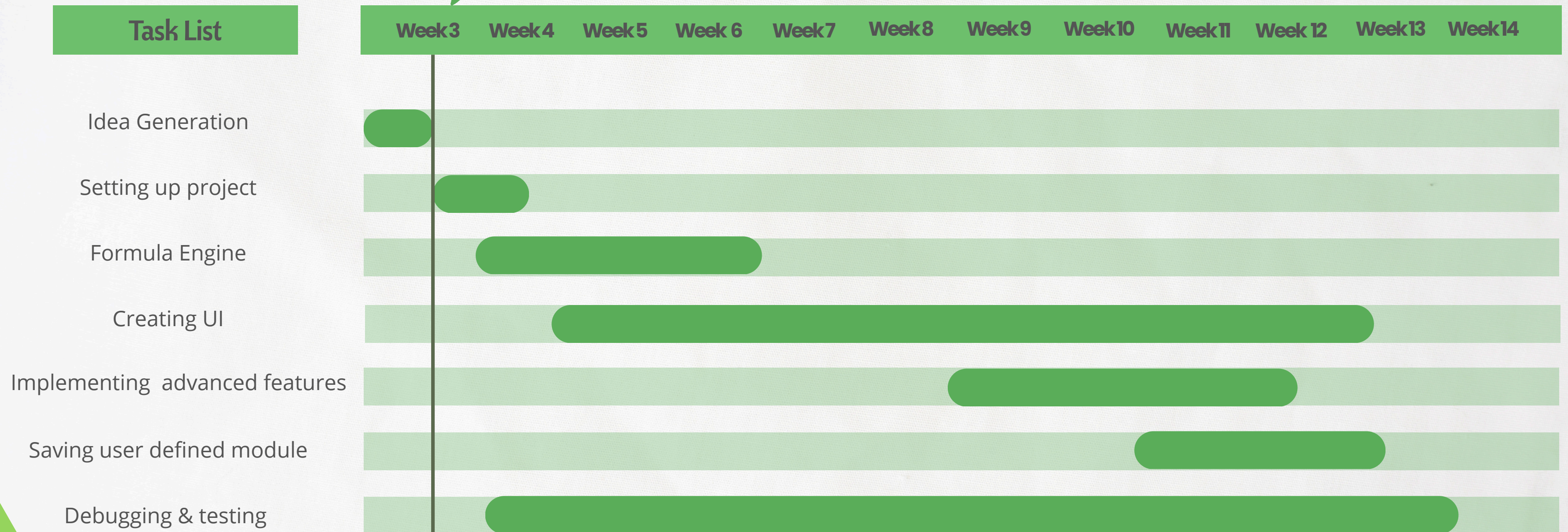


VERSION CONTROL



TIMELINE

We are here



THANK YOU

Please feel free to ask any questions